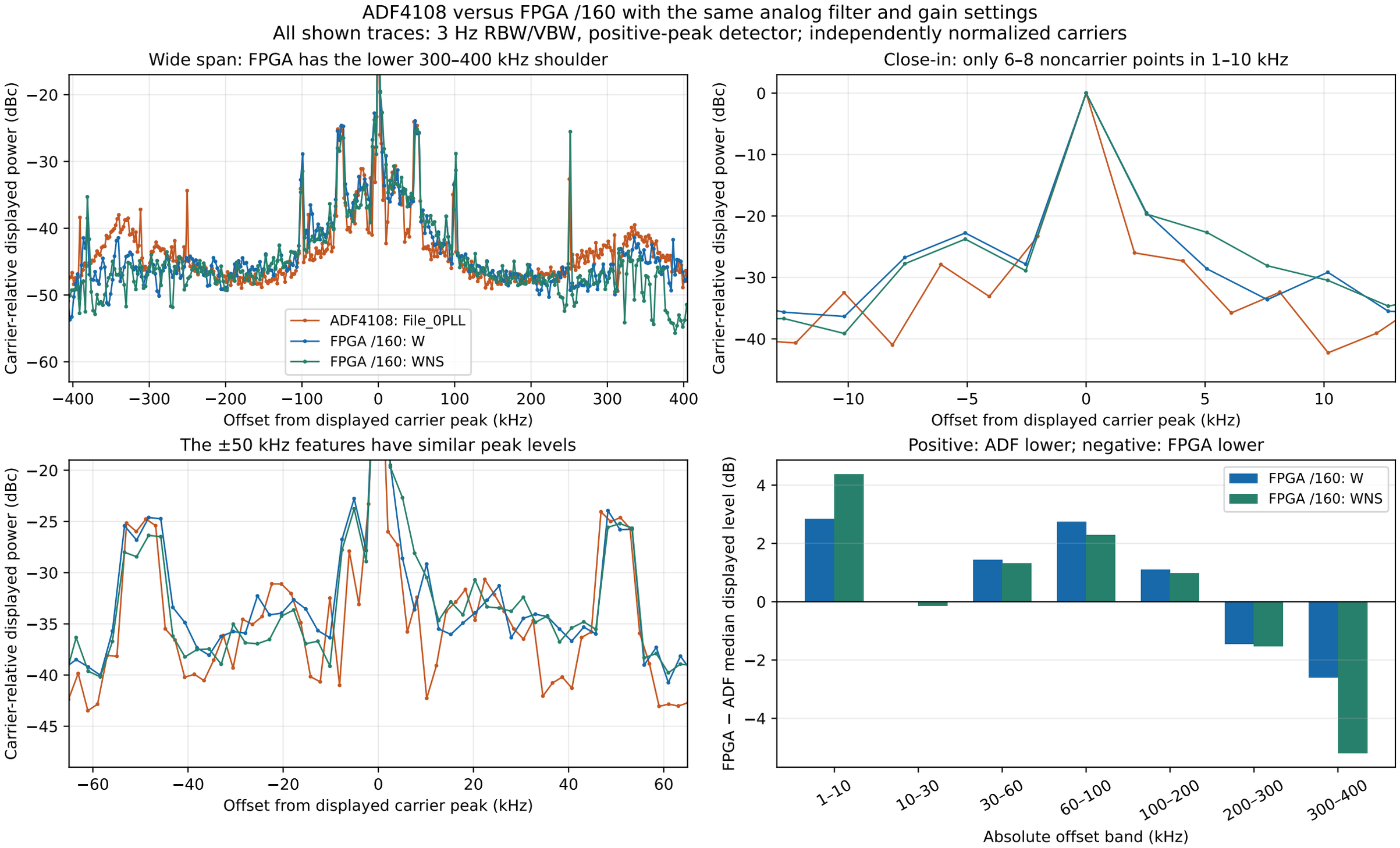

ADF4108 versus FPGA /160 with the same analog filter and gain settings
All shown traces: 3 Hz RBW/VBW, positive-peak detector; independently normalized carriers
Wide span: FPGA has the lower 300–400 kHz shoulder
Carrier-relative displayed power (dBc)
Offset from displayed carrier peak (kHz)
ADF4108: File_0PLL
FPGA /160: W
FPGA /160: WNS
Close-in: only 6–8 noncarrier points in 1–10 kHz
Carrier-relative displayed power (dBc)
Offset from displayed carrier peak (kHz)
The ±50 kHz features have similar peak levels
Carrier-relative displayed power (dBc)
Offset from displayed carrier peak (kHz)
Positive: ADF lower; negative: FPGA lower
FPGA − ADF median displayed level (dB)
FPGA /160: W
FPGA /160: WNS
1–10
10–30
30–60
60–100
100–200
200–300
300–400
Absolute offset band (kHz)
Display spacing: ADF 2.04 kHz, FPGA 2.54 kHz. Band medians are display statistics, not integrated phase noise.
All files carry an incomplete-sweep warning; amplitude and phase noise are not separated.